Unit 13: Long -Run Production

Two Variable Inputs: Isoquants, Substituting Inputs

Objectives of the Study

By studying this unit, the student will be able to

-understand long run production function

Production theory explains how firms transform inputs into outputs. In earlier analysis of short-run production, one factor—usually capital—was assumed fixed, while labor was variable. But, in the long run, all factors of production are variable. This fundamental difference changes the nature of decision-making for firms. When both labor and capital are variable, firms gain flexibility in choosing input combinations to produce a given level of output. In the long run, firms can substitute labor for capital or capital for labor while maintaining the same output level. This flexibility resembles consumer behaviour in utility theory, where consumers substitute between goods while maintaining the same level of satisfaction. In production theory, this substitution is represented graphically using isoquants.

In the long run, both labor (L) and capital (K) are variable. A firm can change the number of workers, the number of machines, factory size, or technology. Because both inputs can vary, the firm can choose different combinations of labor and capital to produce the same level of output. Typically, a firm can produce in a number of different ways, some of which require more labor than others. With both factors variable, a firm can usually produce a given level of output by using a great deal of labor and very little capital, a great deal of capital and very little labor, or moderate amounts of both. That is, the firm can substitute one input for another while continuing to produce the same level of output, in much the same way that a consumer can maintain a given level of utility by substituting one good for another.

Table 13.1: Two Variable Inputs

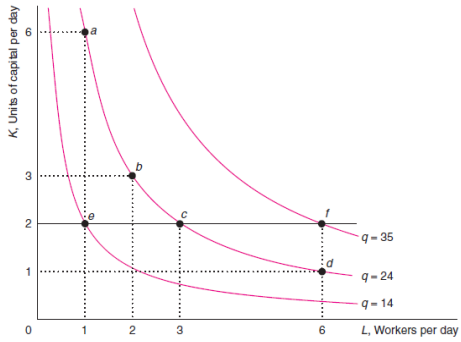
Labour / Capital	1	2	3	4	5	6
1	10	14	17	20	22	24
2	14	20	24	28	32	35
3	17	24	30	35	39	42
4	20	28	35	40	45	49
5	22	32	39	45	50	55
6	24	35	42	49	55	60

The table 13.1 shows the amount of output per day the firm produces with various combinations of labor per day and capital per day. The labor inputs (L) are along the top of the table, and the capital inputs (K) are in the first column. The table shows four combinations of labor and capital that the firm can use to produce 24 units of output: The firm may employ (a) 1 worker and 6 units of capital, (b) 2 workers and 3 units of capital, (c) 3 workers and 2 units of capital, or (d) 6 workers and 1 unit of capital. These combinations produce the same output but use different input proportions.

Isoquants

An isoquant is a curve that shows all efficient combinations of labor and capital that produce the same level of output. If the production function is q = f (L, K). Then an isoquant for output level $\bar{q}$ bar is defined as: $\bar{q}$ = f (L, K). Each point on the isoquant represents an efficient combination of inputs yielding the same output. Isoquants provide a graphical representation of the firm's flexibility in input choice. An isoquant shows the flexibility that a firm has in producing a given level of output.

Figure 13.1: Isoquant



In the figure 13.1 we can see three isoquants corresponding to three levels of output. These isoquants are smooth curves because the firm can use fractional units of each input. We can use these isoquants to illustrate what happens in the short run when capital is fixed and only labor varies. If the firm holds one factor constant and varies another factor, it moves from one isoquant to another. In contrast, if the firm increases one input while lowering the other appropriately, the firm stays on a single isoquant. In long-run production analysis, isoquants help us understand how firms adjust input combinations while maintaining or changing output levels. The key distinction is between movement along an isoquant and movement from one isoquant to another. If a firm increases one input while decreasing the other in the correct proportion, it moves along the same isoquant. This means output remains constant. The firm is simply substituting one input for another without changing the level of production. This

movement reflects the firm's flexibility in choosing different input combinations to produce the same quantity of output. In contrast, if the firm increases one input while keeping the other input constant, it moves to a higher isoquant, which represents a higher level of output. Since both inputs are not being adjusted in opposite directions to keep output constant, total production increases. This shift indicates that the firm has moved to a new isoquant representing a greater quantity of output. Thus, movement along an isoquant shows substitution between inputs with no change in output, while movement from one isoquant to another reflects a change in output level due to increased overall input usage. Understanding this distinction is essential for analysing long-run production decisions and cost minimization behaviour.

Properties of Isoquants

Isoquants are an important tool in production theory. Just as indifference curves in consumer theory, isoquants show combinations of inputs that produce the same level of output. However, instead of holding utility constant, isoquants hold output constant.

1. Isoquants Further from the Origin Represent Higher Output: Isoquants that lie farther from the origin represent higher levels of output. This is because moving away from the origin means using more inputs (labor and capital), assuming efficient production. This property reflects a basic economic principle: when additional inputs are used efficiently, output increases. Thus, isoquants positioned farther from the origin correspond to larger production levels.

2. Isoquants Do Not Intersect: Isoquants can never intersect each other. If two isoquants were to cross, it would imply that a single combination of labor and capital produces two different levels of output at the same time, which is logically impossible. Intersection would also imply technical inconsistency and inefficiency. Hence, isoquants are always separate and distinct.

3. Isoquants Are Downward Sloping: Isoquants are negatively sloped (downward sloping from left to right). This property reflects the principle of input substitution. If a firm reduces the quantity of one input, it must increase the quantity of the other input to maintain the same level of output. If an isoquant were upward sloping, it would imply that the firm could increase both labor and capital and still produce the same output. This would be inefficient because using more inputs should increase output, not keep it constant. Therefore, isoquants must slope downward to show the trade-off between inputs while holding output constant.

4. Isoquants Are Thin (Represent Efficient Combinations Only): Isoquants are thin curves because they represent only efficient combinations of inputs. An efficient input combination means the firm cannot reduce one input without increasing the other. There is no waste in

production. Input combinations that lie above an isoquant are inefficient because the same output could be produced using fewer inputs. Resources are being wasted. Isoquants include only technically efficient input combinations, and inefficient points are excluded.

Shape of Isoquants

The shape of an isoquant shows how easily one input can be substituted for another while keeping output constant. In other words, the curvature reflects the degree of substitutability between inputs. Depending on the nature of the production process, isoquants may be straight lines, L-shaped, or smoothly convex.

1. Perfect Substitutes: When two inputs can completely replace each other at a constant rate, they are called perfect substitutes. In this case, isoquants are straight lines with a constant slope. Example Production Function: $q = x + y$. Here, output depends only on the total number of input units, regardless of the combination. This means: 1 unit of x produces 1 unit of output. 1 unit of y produces 1 unit of output. 0.5 units of x and 0.5 units of y together produce 1 unit of output. Since one unit of x can always replace one unit of y, the marginal rate of technical substitution (MRTS) is constant. The slope of the isoquant is -1, meaning the firm can substitute one unit of x for one unit of y without affecting output.

2. Fixed-Proportions Production Function (Perfect Complements). In some production processes, inputs must be used in fixed ratios. In such cases, isoquants are L-shaped (right-angled). This is known as a fixed-proportions production function or perfect complements. For Example, producing a cereal package requires: one box and one unit of cereal. If a firm has 5 boxes and 5 cereal units then it can produce 5 packages. If 10 cereal units but only 5 boxes then it can still produce only 5 packages. Extra cereal without additional boxes adds nothing to output.

3. Typical Case: Convex Isoquants: In most real-world production processes, isoquants are smooth and convex to the origin. This reflects the principle of diminishing marginal rate of technical substitution (MRTS). As the firm substitutes labor for capital at first, labor can easily replace capital. But as more labor is added and capital is reduced, each additional unit of labor becomes less effective at replacing capital. Therefore, the MRTS diminishes as we move down along the isoquant. The isoquant becomes flatter. Suppose a firm reduces capital and increases labor initially, when capital is abundant, labor can replace it easily. Later, when capital becomes scarce, it becomes harder for labor to compensate for further reductions in capital. Thus, the trade-off rate declines.

Marginal Rate of Technical Substitution (MRTS)

The Marginal Rate of Technical Substitution (MRTS) refers to the slope of an isoquant. It shows the rate at which one input can be substituted for another while keeping output constant. Mathematically, MRTS = ΔK/ΔL. It measures how much capital (K) can be reduced when one additional unit of labor (L) is added, without changing the level of output. MRTS shows the trade-off between inputs. Isoquants are downward sloping because if labor increases, capital must decrease to keep output constant. If capital increases, labor must decrease. Since ΔK is negative when ΔL is positive, the MRTS is negative. However, in practical analysis, economists often focus on the absolute value of MRTS and ignore the negative sign, because the sign simply reflects the downward slope of the isoquant. MRTS can also be expressed using marginal products

$$MRTS = - MPL/MPK$$

Where,

MPL = Marginal Product of Labor and

MPK = Marginal Product of Capital.

This means MRTS depends on the productivity of each input.

Diminishing MRTS

In most real-world production processes, MRTS diminishes as we move down along an isoquant. This is because when capital is abundant and labor is scarce, labor can easily replace capital, as more labor is added and capital is reduced, it becomes harder for labor to substitute for capital and therefore, the amount of capital that can be given up for each extra unit of labor decreases. This phenomenon is called the law of diminishing MRTS, and it explains why isoquants are convex to the origin.

Cobb–Douglas Production Function

The Cobb–Douglas production function is one of the most widely used production functions in economics. It describes how output depends on the quantities of labor and capital used in production. General Form

$$q = AL^{\alpha}K^{\beta}$$

Where,

A = Technology parameter (Total Factor Productivity),

L = Quantity of labor,

K = Quantity of capital,

α = Output elasticity of labor and

β = Output elasticity of capital.

A – Technology Parameter measures the level of technology or efficiency in production. A higher value of A means the firm can produce more output with the same inputs. Improvements in technology increase A.

α – Output Elasticity of Labor: If α = 0.5, then a 1% increase in labor increases output by 0.5%, holding capital constant. It also shows labour's contribution to production.

β – Output Elasticity of Capital. If β = 0.4, then a 1% increase in capital increases output by 0.4%, holding labor constant.

In the Cobb–Douglas function, marginal products are proportional to average products.

Marginal Products

a) Marginal Product of Labor (MPL).

$MPL = \alpha \frac{q}{L}$

Since, APL = q/L.

Then, MPL = α APL.

This means, Marginal product of labor is α times average product of labor.

If α < 1, then MPL < APL. As labor increases, MPL eventually diminishes.

b) Marginal Product of Capital (MPK)

$MPK = \beta \frac{q}{K}$

Since, APK = q/K.

Then, MPK = β APK.

Thus, MPK is proportional to APK.

If β < 1, capital also exhibits diminishing marginal returns.

This proportional relationship is a special and convenient feature of the Cobb–Douglas function.

MRTS in Cobb–Douglas Function

We know that MRTS = -MPL/MPK. Substituting MPL and MPK:

$$MRTS = -\frac{\alpha \frac{q}{L}}{\beta \frac{q}{K}}$$

Since q appears in both numerator and denominator, it cancels out.

Then, MRTS = – (α / β) (K / L). So, we can say that MRTS depends on the ratio of the elasticities (α/β) and the capital–labor ratio (K/L).

Long-run production analysis with two variable inputs provides deeper insight into how firms substitute between labor and capital. Isoquants graphically represent combinations of inputs producing equal output. Their properties—downward slope, convexity, and non-intersection—reflect efficient production. The marginal rate of technical substitution measures input substitutability and equals the negative ratio of marginal products. Diminishing MRTS explains why isoquants are convex. The Cobb–Douglas production function offers a practical framework to derive marginal products and MRTS analytically. For the service firm example, the MRTS depends on the capital-labor ratio and declines as labor increases relative to capital. Overall, long-run production theory equips firms with the tools necessary for efficient resource allocation, cost minimization, and optimal production planning.

Multiple Choice Questions

1. What is the primary difference between short-run and long-run production?

A. In the short run, all production factors are variable.

B. In the long run, all factors of production are variable.

C. In the long run, firms cannot change their production methods.

D. In the short run, firms have complete flexibility in input combinations.

Answer: B. In the long run, all factors of production are variable.

2. What does an isoquant represent in production theory?

A. A curve showing all combinations of inputs that produce the same level of output.

B. A method for measuring total output in the short run.

C. A fixed amount of labor required for production.

D. A representation of diminishing returns in production.

Answer: A. A curve showing all combinations of inputs that produce the same level of output.

3. What does the marginal rate of technical substitution (MRTS) measure?

A. The total output produced by all inputs combined.

B. The rate at which one input can be substituted for another while maintaining the same level of output.

C. The total cost of production for a firm.

D. The productivity of labor in relation to capital.

Answer: B. The rate at which one input can be substituted for another while maintaining the same level of output.

4. In the Cobb-Douglas production function, what does the parameter ‘A’ represent?

A. The total amount of capital used in production.

B. The technology parameter or total factor productivity.

C. The output elasticity of capital.

D. The average product of labor.

Answer: The technology parameter or total factor productivity.

5. What does diminishing MRTS imply in production?

A. Substituting labor for capital always increases output.

B. As more labor is added, it becomes harder to substitute labor for capital.

C. All inputs can be substituted perfectly without loss of output.

D. The total output will always increase with more capital.

Answer: B. As more labor is added, it becomes harder to substitute labor for capital